# xAI 1: Descriptive Interpretability

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# Housekeeping

Please do the exam-time survey, if you haven't yet!

Aasa will send a first version of the exam schedule on Friday.

Please check back: we expect there will be revisions and reschedulings.

# Learning Goals

- Understand the goals of model interpretability and explainable AI.
- Learn about different methods for exploratory & descriptive model interpretabily.
- Differentiate between valid vs invalid conclusions for individual interpretability methods.

# Interpretability and Explainable AI

## Interpretion:

- Goal: Understand the model
- Post-hoc analysis
- Audience: researchers, developers
- "mechinterp": mechanistic interpretability

## Explanation:

- Goal: Understand the model output/response
- "Explainable-by-design": Model architecture creates explanations
- and also: post-hoc explanations
- Audience: users (in theory)

# Interpretability and Explainable AI

## ISO definitions

**3.1.42**
**interpretability**
<AI (3.1.13)> level of understanding how the underlying (AI) technology works

**3.1.31**
**explainability**
<AI (3.1.13)> level of understanding how the AI-based system (3.1.9) came up with a given result

Interpretability and explainability overlap, interesect, and reinforce each other

# Explainable AI

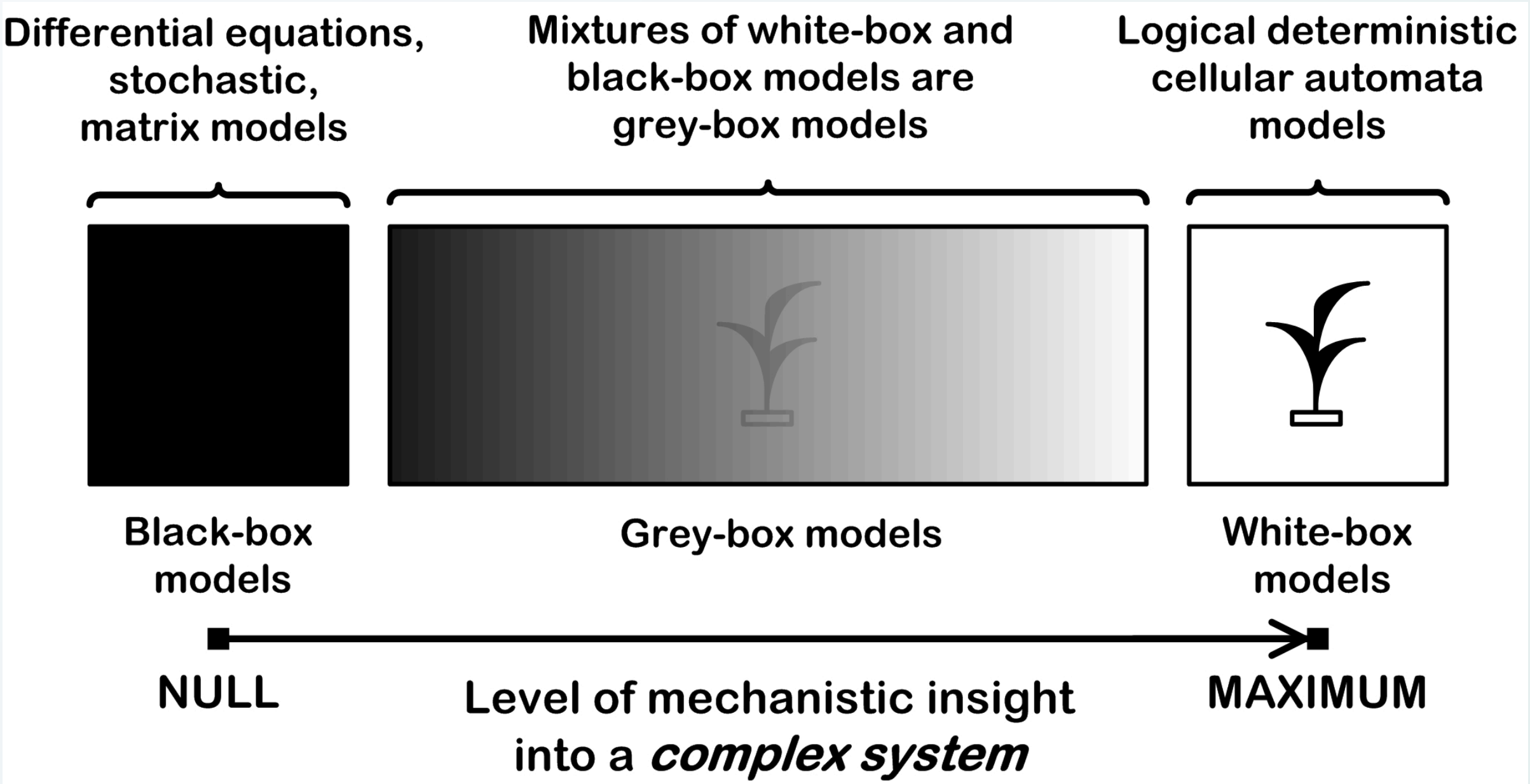
## Legal: "Right to an explanation"

GDPR

> (h) the existence of automated decision-making, including profiling, referred to in Article 22(1) and (4) and, at least in those cases, meaningful information about the logic involved, as well as the significance and the envisaged consequences of such processing for the data subject.

## Engineering: "glass box" models

- Explainable-by-design
- or tools to make black box more transparent?

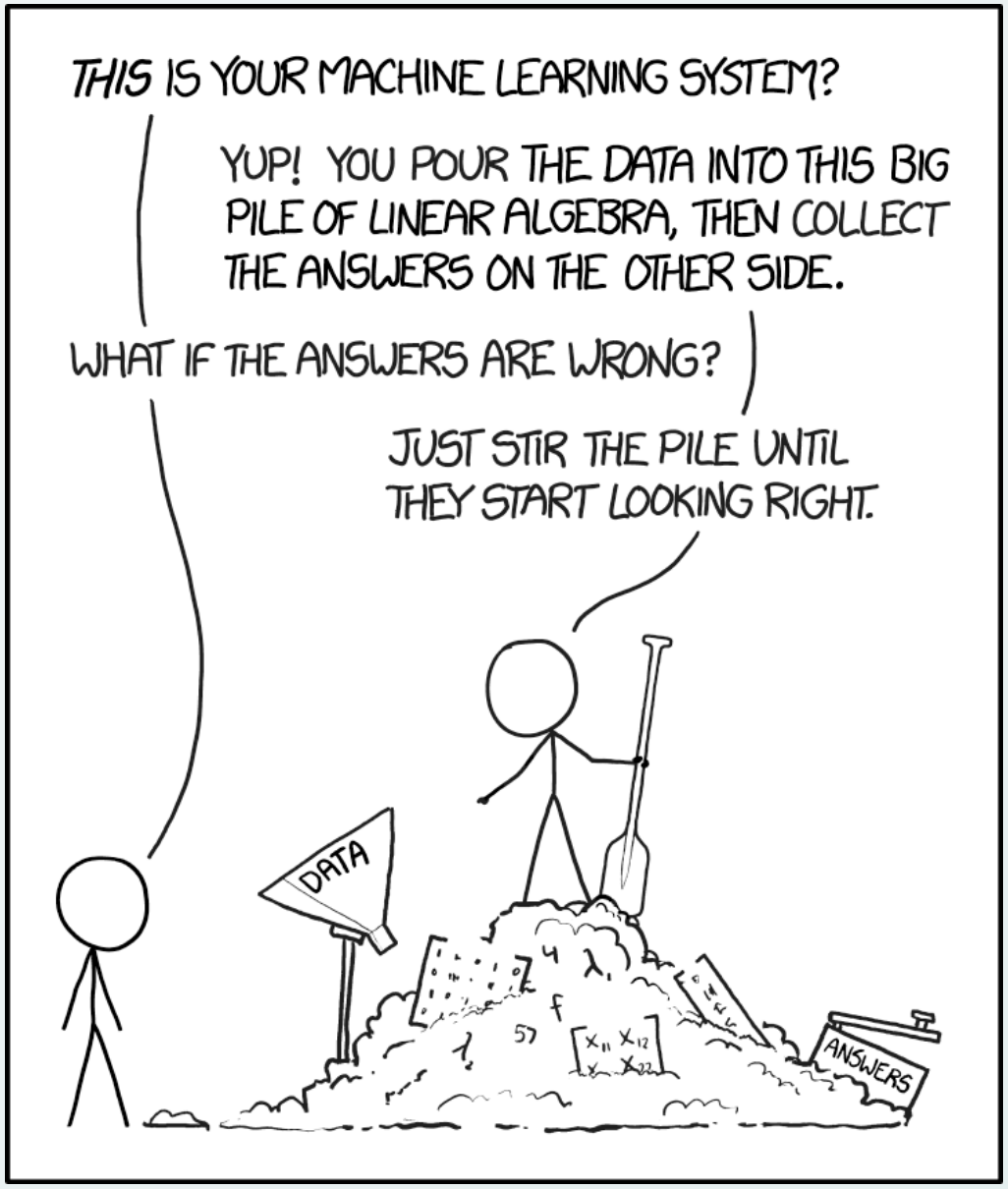



DOI: 10.7717/peerj.948/fig-1

# Interpretability

Mechanistic interpretability: trying to look inside the "black box" of large NN models.

What are the billions of parameters doing?



# "Why?" and "How?"

**Explainability**: Why did the model predict this diagnosis?

- Was it because of medical features or shortcut learning?

**Interpretability**: How does the model predict diagnoses?

- Does it have a good representation of the medical features?
- or is it giving too much weight to shortcut features?

Again, in practice, this distinction is not always sharp, or critical - interpretability methods can help provide explanations, and explanations help with interpretability.

Towards A Rigorous Science of Interpretable Machine Learning. Doshi-Velez & Kim (2017)

# Question - discussion

- Have you worked with interpretability or explainable AI methods before?
- Have you debugged AI models by looking at their internals?

# Roadmap: Upcoming lectures

Today (8.10): **Descriptive** methods for interpretability

22.10: **Causal** methods for interpretability

29.10: Feature relevance methods (interp/explainability)

5.11: Intrinsically explainable models & validating XAI part 1

12.11: Sample influence methods & validating XAI part 2

19.11: Uncertainty

# Today: Descriptive interpretability

Describing and observing model structure, e.g. learned representation space

**Methodologies**

1. Representation Space Visualizations
2. Neighborhood analyses
3. Linear probing
4. Un-embedding lenses
5. Attention flows

Q: Are any of these familiar to you?

# Representation Space Analyses

Representations aka embeddings: vector representation of input $e = f(x)$

## Warning!

- For non-mathematicians, this is going to be tougher than the previous material.
  - You may have to spend a long time studying the next two parts.
- If you are not used to thinking about hyper-planes in high-dimensional spaces, now is the time to learn.
- To deal with hyper-planes in a 14-dimensional space, visualize a 3-D space and say "fourteen" to yourself very loudly. Everyone does it.
  - But remember that going from 13-D to 14-D creates as much extra complexity as going from 2-D to 3-D.

Geoffrey Hinton

# Visualizing Representation Space

Methods: PCA, t-SNE, **UMAP** (UMAP tutorial, UMAP documentation)

"UMAP constructs a high dimensional graph representation of the data,
then optimizes a low-dimensional graph to be as structurally similar as possible."

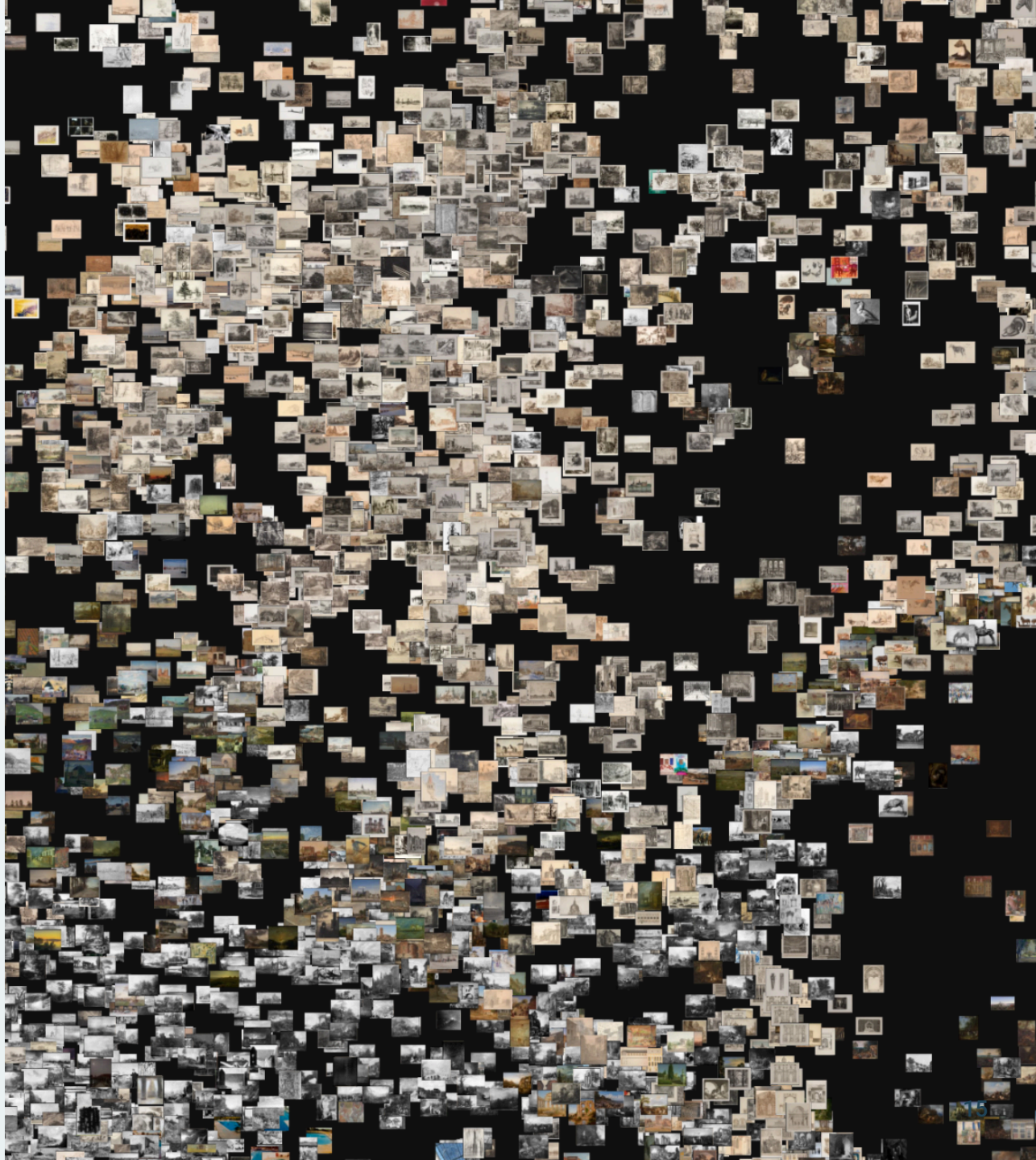
0. Distance metric: e.g. Euclidean, cosine
1. Space → graph: discretization step
    - `n_nearest` parameter: how connected to make this graph
    - small vs large neighborhoods = show local vs global structure
2. Flattening step: Original to reduced dimensions

# What to conclude from UMAP visualization?

**4-minute discussion:**

1. Is UMAP showing only linear or also non-linear relationships?
   (What about PCA?)
2. What does proximity tell us?
3. What does distance tell us?
4. What do clusters tell us?
5. What do axes tell us?

UMAP of SMK collection: pixplot.smk.dk



# What to conclude from UMAP visualization?

- Highly similar points are near each other: proximity indicates similarity

## What can we *not* conclude?

- Distances are not meaningful
- Cluster separation depends strongly on hyperparameters
- Axes are not meaningful (unlike PCA!)
- Distant points could be similar in representation *sub*-space

## UMAP is for exploratory data analysis - not for quantitative conclusions.

- Biologists, stop putting UMAP plots in your papers

# Neighborhood analyses

Which points lie close together in representation space?

Nearest neighbors via a distance metric (cosine, Euclidean) - just like UMAP

Possible Questions:

1. Single space: Do nearest neighbors correspond to intuition, gold categories?
    - Word embeddings: synonyms as nearest neighbors
    - Contextual word embeddings: word senses

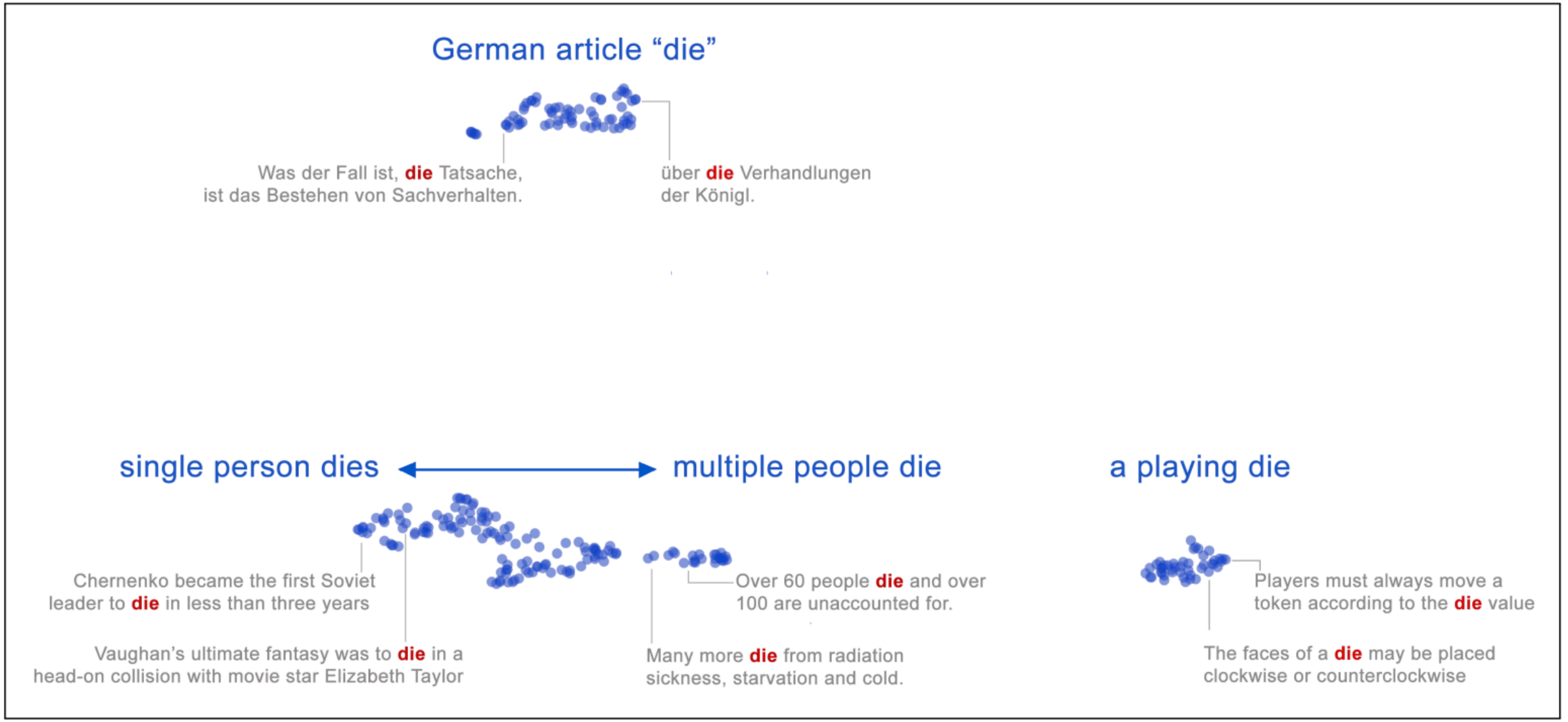



**Figure 10.2** Each blue dot shows a BERT contextual embedding for the word *die* from different sentences in English and German, projected into two dimensions with the UMAP algorithm. The German and English meanings and the different English senses fall into different clusters. Some sample points are shown with the contextual sentence they came from. Figure from Coenen et al. (2019).

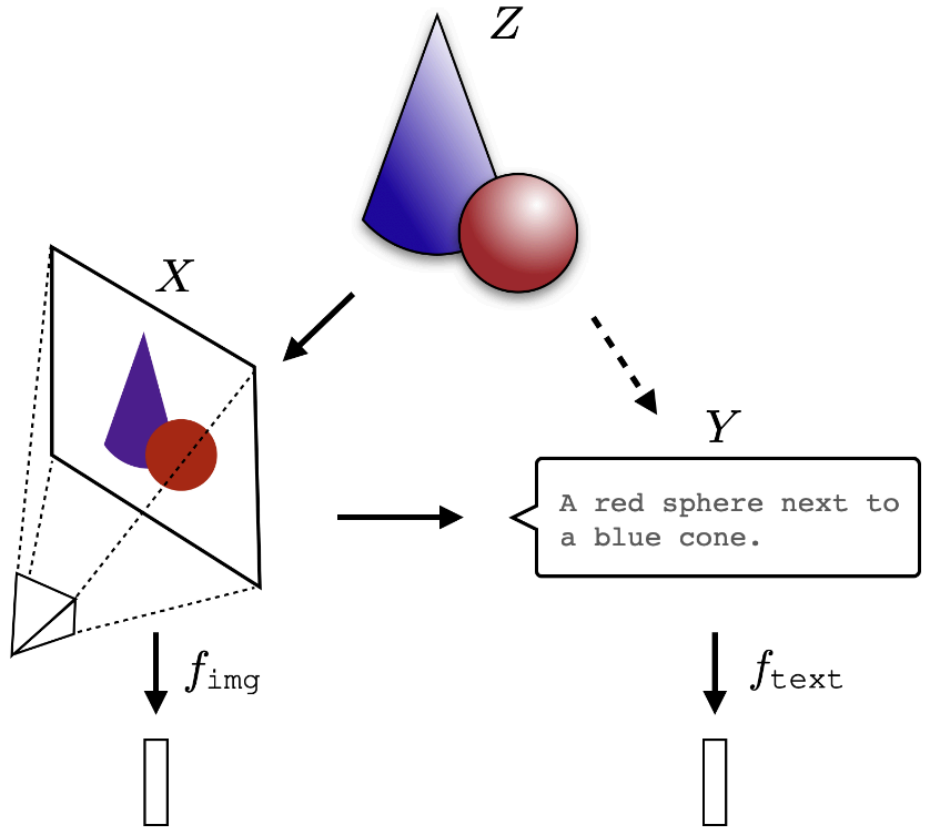
# Neighborhoods: PRH

2. Alignment between spaces:
   Are two spaces structured similarly - do they have similar nearest-neighbor patterns?

Huh et al., ICLR 2024. Position: The Platonic Representation Hypothesis

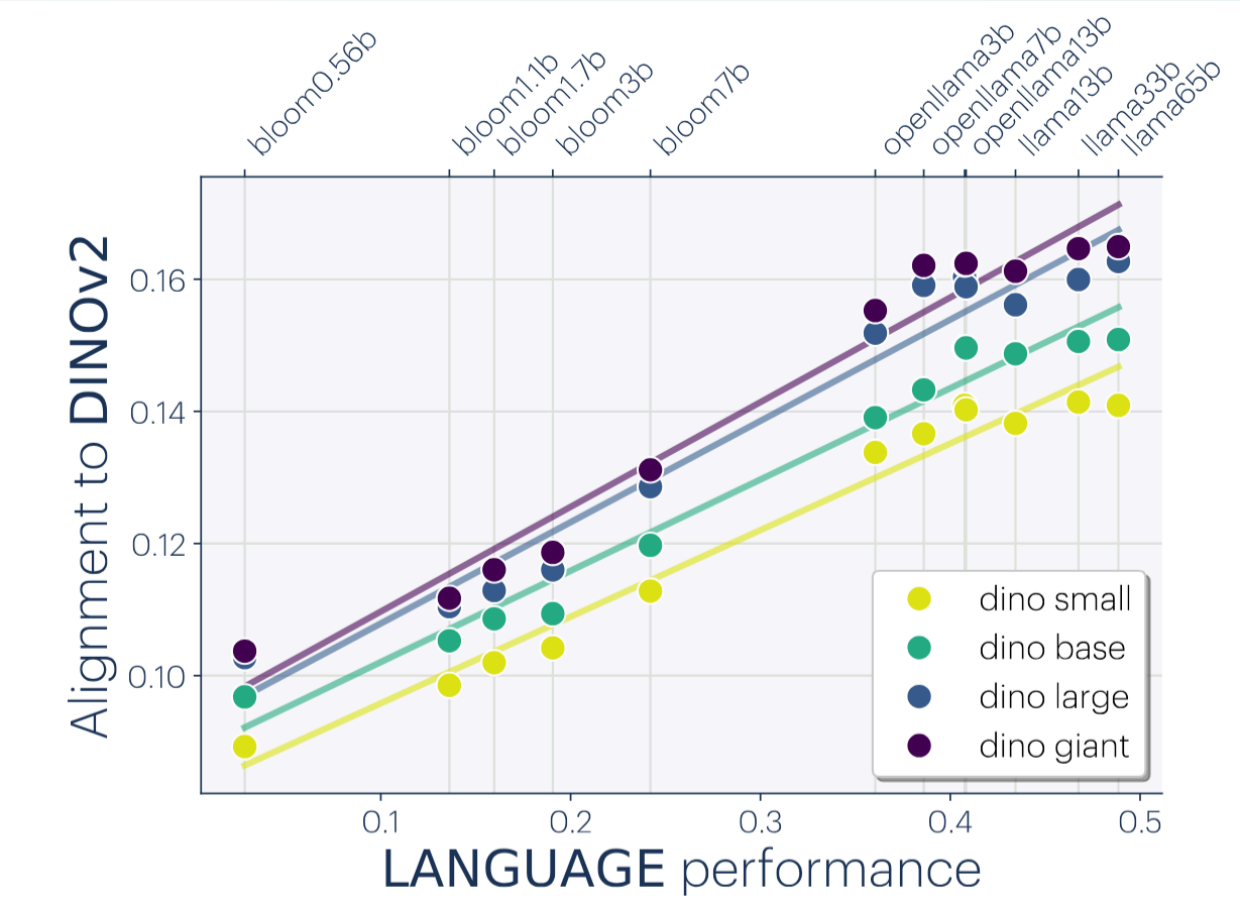
**The Platonic Representation Hypothesis**

Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.



# Neighborhoods: PRH

Mutual nearest neighbors on 1k samples from image-caption dataset (WIT)




Huh et al., ICLR 2024. Position: The Platonic Representation Hypothesis

# Neighborhoods: Back into Plato's Cave

BUT: measuring alignment by NNs depends on number of potential neighbors

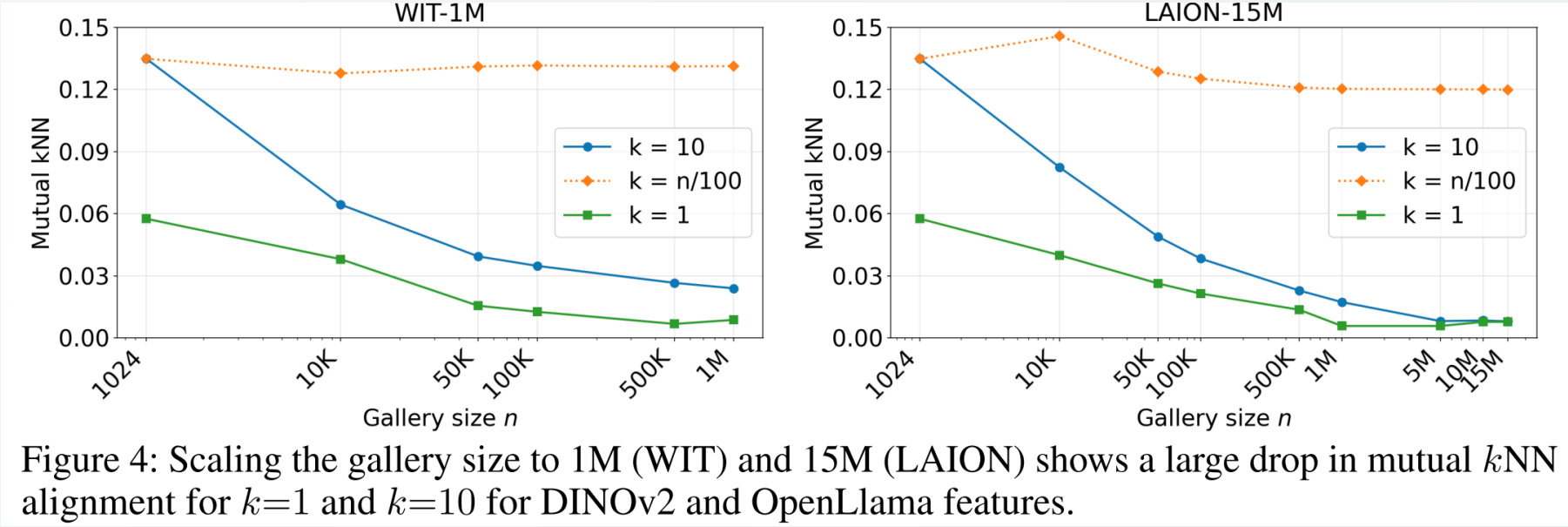



Figure 4: Scaling the gallery size to 1M (WIT) and 15M (LAION) shows a large drop in mutual $k$NN alignment for $k$=1 and $k$=10 for DINOv2 and OpenLlama features.

Alignment disappears with higher neighborhood density!

Koepke et al., 2026. Back into Plato's Cave: Examining Cross-modal Representational Convergence at Scale

# What to conclude from nearest neighbors?

- Local, not global, structure

Conclusions are dependent on:

- Hyperparameters (e.g. $k$)
- Distance metric
- Neighborhood density
- Hubness - hyperpopular neighbors (Radovanovic et al., 2010)

# Aside: "Curse of dimensionality"

Issues with distances in high-dimensional space:

- **Distance concentration**: Distances in high-dimensional space tend to become equal, with randomly distributed data
- **Anisotropy** in representation space: If representations are not evenly distributed ("not isotropic"), similarity will be high on average, i.e. expected cosine similarity will be $> 0$. (Ethayarajh, 2019, Mimno & Thompson 2017)
- **Rogue dimensions**: a small number of very large magnitude dimensions can dominate the cosine similarity measure (Timkey and van Schijndel 2021)

# Aside: "Curse of dimensionality"

## Sometimes helpful: z-score standardization

Given mean $\mu$ and per-dimension standard deviation $\sigma$ (estimated from from a dataset), transform vector $x$:

$$z = \frac{x - \mu}{\sigma}$$

Questions?

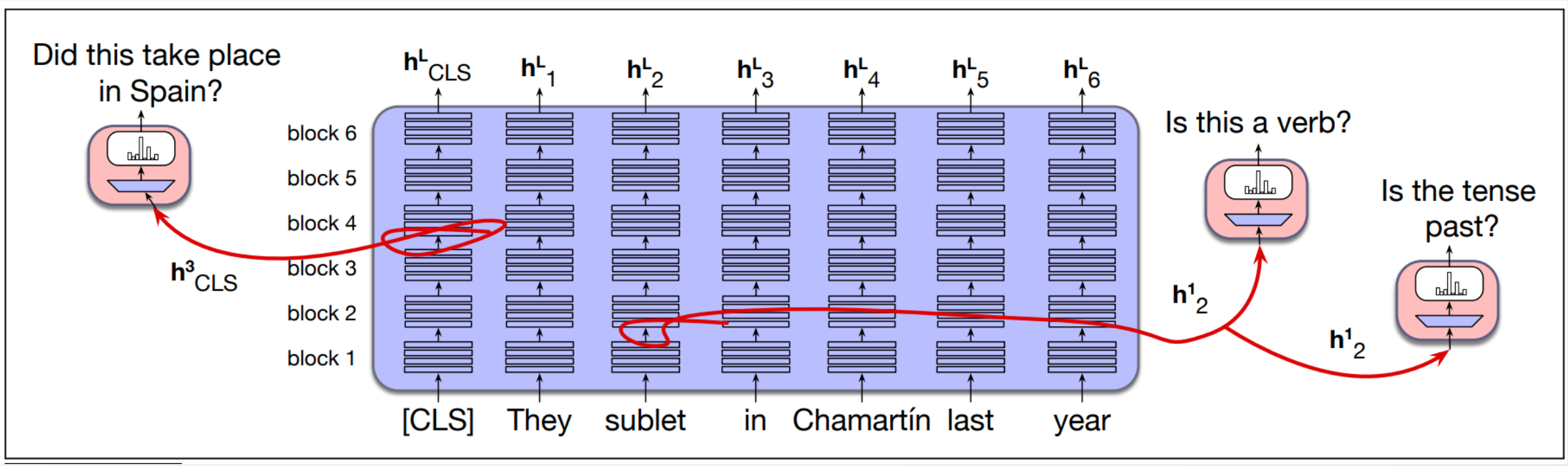
Break!

# Probing: Motivation

- UMAP, NNs etc. using cosine sim etc. look at the **whole** space, all dimensions.
- Sub-spaces (selection of dimensions) can also be informative about specific features.
    - e.g. "enjoys death metal" might be a sub-space, not the main feature driving similarity.

How to find the subspace responsible for a given feature? Learn it!

- Learn a minimal linear map - a **probe** from representation to feature classifier
- Less minimal maps (MLPs) are also possible - but tell you less about representation




Speech and Language Processing, 3rd Edition. Jurafsky & Martin, 2026

# Probing: Methodology

Probing has two phases, training and testing. Training a probe has 4 steps:

1. Choose a property we want to examine (like part of speech, or some aspect of meaning, or which entity a pronoun corefers with)
2. Create a training and test set of embeddings at some position and level we are interested in studying (like "the output of the feedforward network at transformer layers 2 through 16"), labeled with the correct answer (the relevant part of speech, or tense, or whatever).
3. Pass the training set sentences through the LLM, and extract activations
4. Now freeze the weights of the LLM, and train only the weights of the classifier to map from an embedding to a property.

Speech and Language Processing, 3rd Edition. Jurafsky & Martin, 2026

# Probing: Layer-wise behavior

Early layers: word-level tasks (POS, Entities)

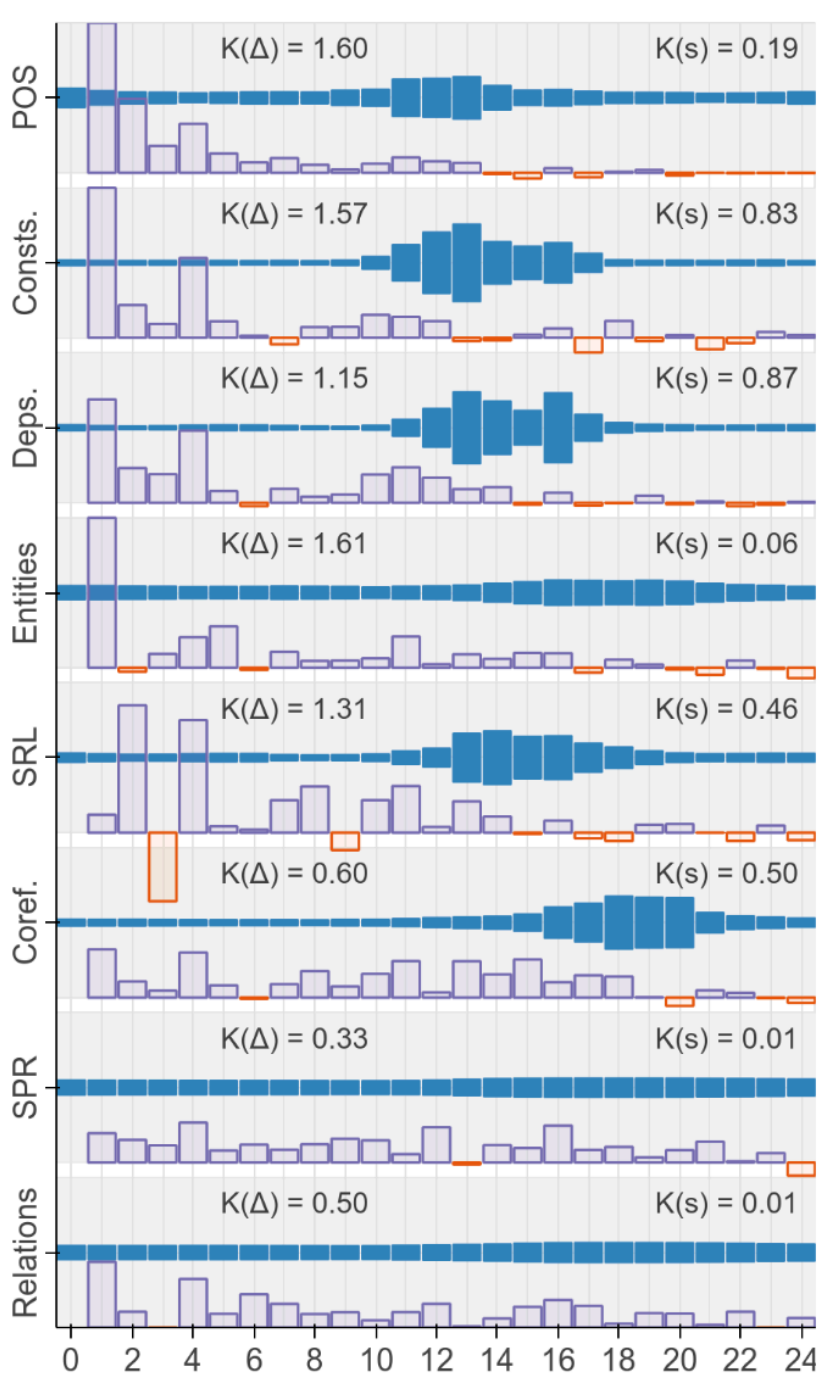
Later layers: sentence-level tasks (Syntax/Deps, Coreference)

Semantic tasks (relations, srl) across layers

Solid Blue: Mixing weights - important layers for the task

Purple: Improvement in probing up to layer L-1 when adding L

BERT Rediscovers the Classical NLP Pipeline (Tenney et al., ACL 2019)

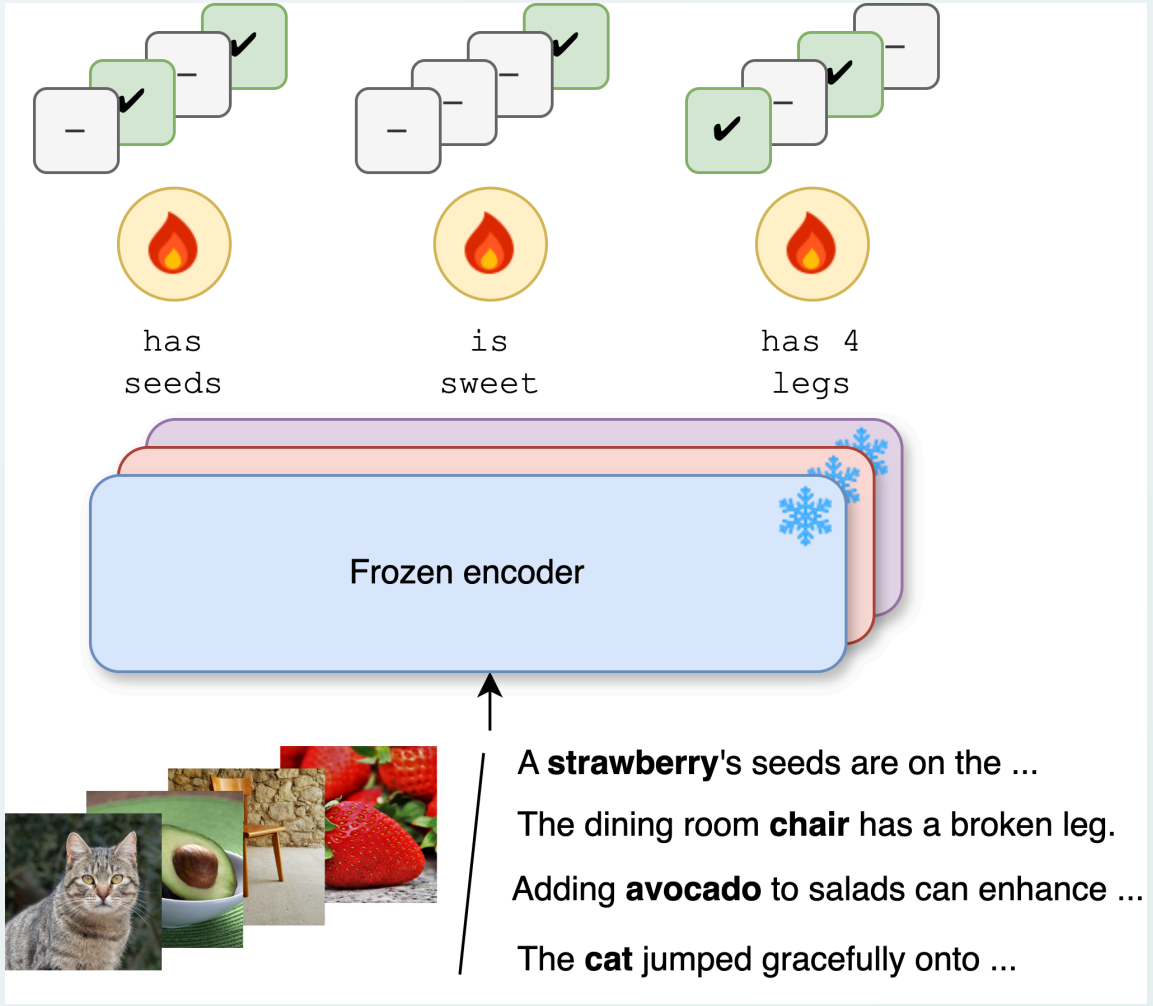


# Probing: Attributes in Language and Vision

*Semantic norms* as concept attributes:

```
avocado: is_green, has_seed
strawberry: is_red, has_seed, is sweet
cat: has_4_legs, has_tail, has_fur
```

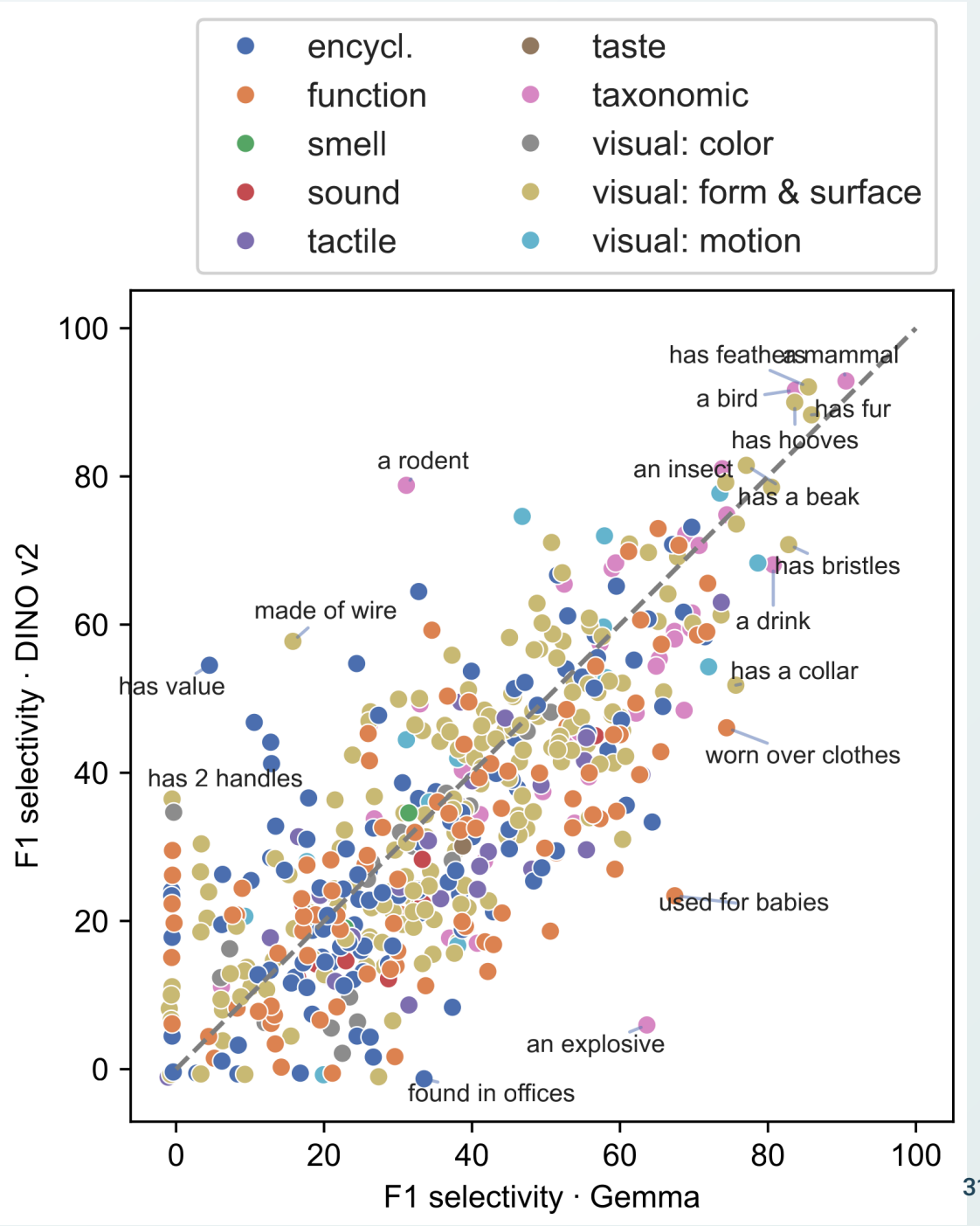
Seeing What Tastes Good: Revisiting Multimodal Distributional Semantics in the Billion Parameter Era (Oneata et al., Findings ACL 2025)



# Probing: Attributes in Language and Vision

**Result:** Probe performance is similar across vision & language encoders.

Seeing What Tastes Good: Revisiting Multimodal Distributional Semantics in the Billion Parameter Era (Oneata et al., Findings ACL 2025)



# What can probing tell you? What can't it?

Discuss!

# What can probing tell you?

If probe works: feature information is linearly-encoded in representation.

- Global organization, not local neighborhood.

But: Probe itself can also learn a lot! *Control* comparisons are important (Belinkov, 2022)

- e.g. Random control task (Hewitt & Liang, 2019)

## What can't it tell you?

Doesn't tell us model is *using* the information: correlation, not causation.

If linear probe doesn't work: feature could still be non-linearly encoded in representation.

## Q: What kind of features might be non-linear?

# Non-linear representations

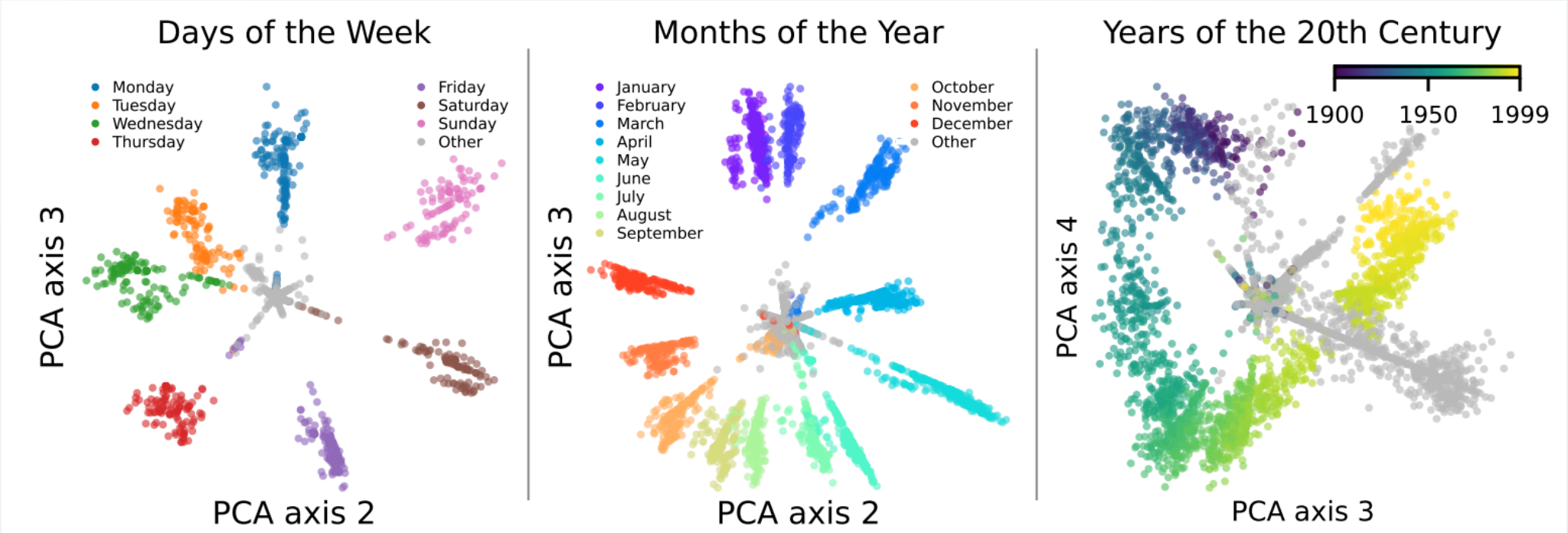



Figure 1: Circular representations of days of the week, months of the year, and years of the 20th century in layer 7 of GPT-2-small colored by the token they fire on. These representations were discovered via clustering SAE dictionary elements, described in Section 4. Points are colored according to the token which created the representation. See Fig. 14 for other axes and Fig. 15 for

Engels et al., ICLR 2025 Not All Language Model Features Are One-Dimensionally Linear

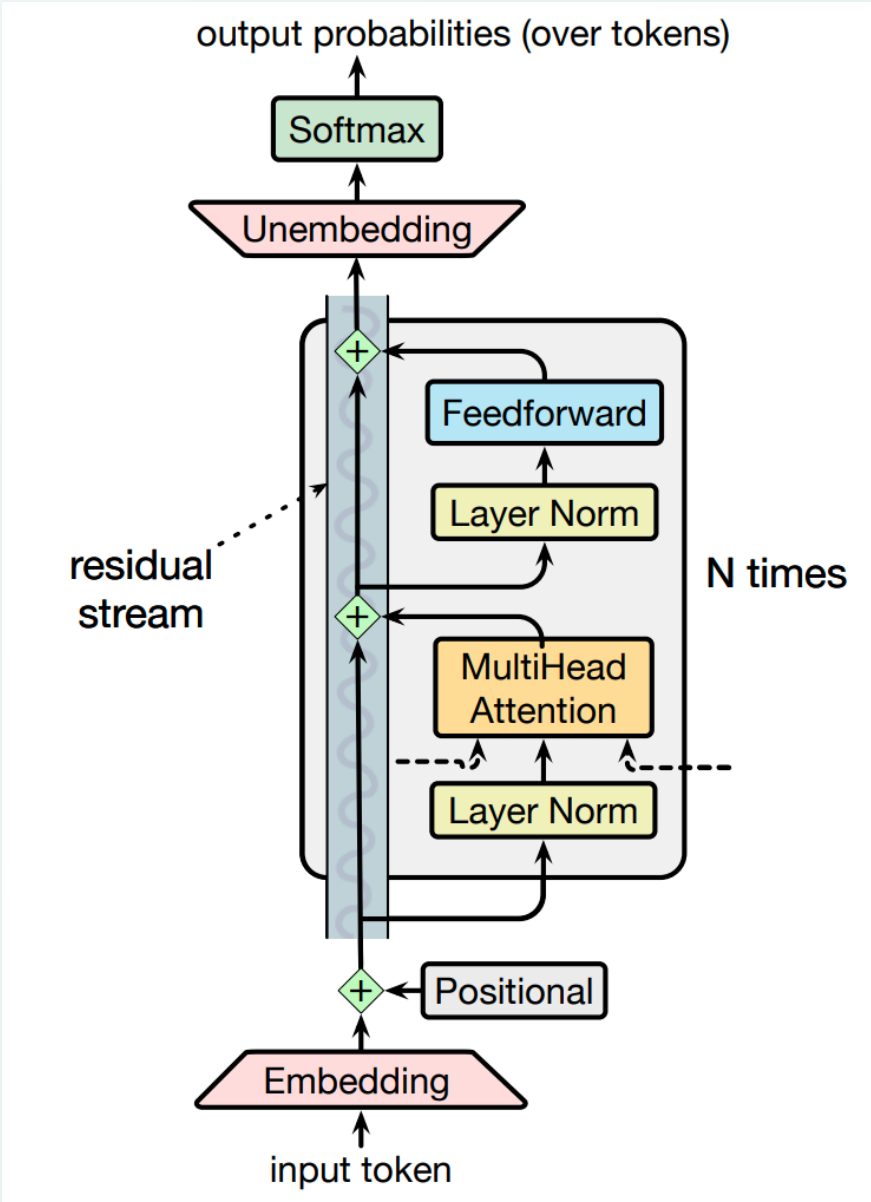
# Logit Lens

Idea: apply un-embedding at intermediate layers (after feed-foward) to "decode" representation back into token vocabulary.

Surprisingly this (kind of) works!

nostalgiabraist: logitlens

illustrated transformer



# Logit Lens, improved

- Tuned Lens, Belrose++2025: Learn a layer-wise map to final pre-embedding layer
- J-lens, Gurnee++2026: Similar map based on Jacobian derivative (not learned)
- Patchscope, Ghandeharioun++2024: generalization
- LatentLens, Krojer++2026 for interpreting visual tokens in VLMs

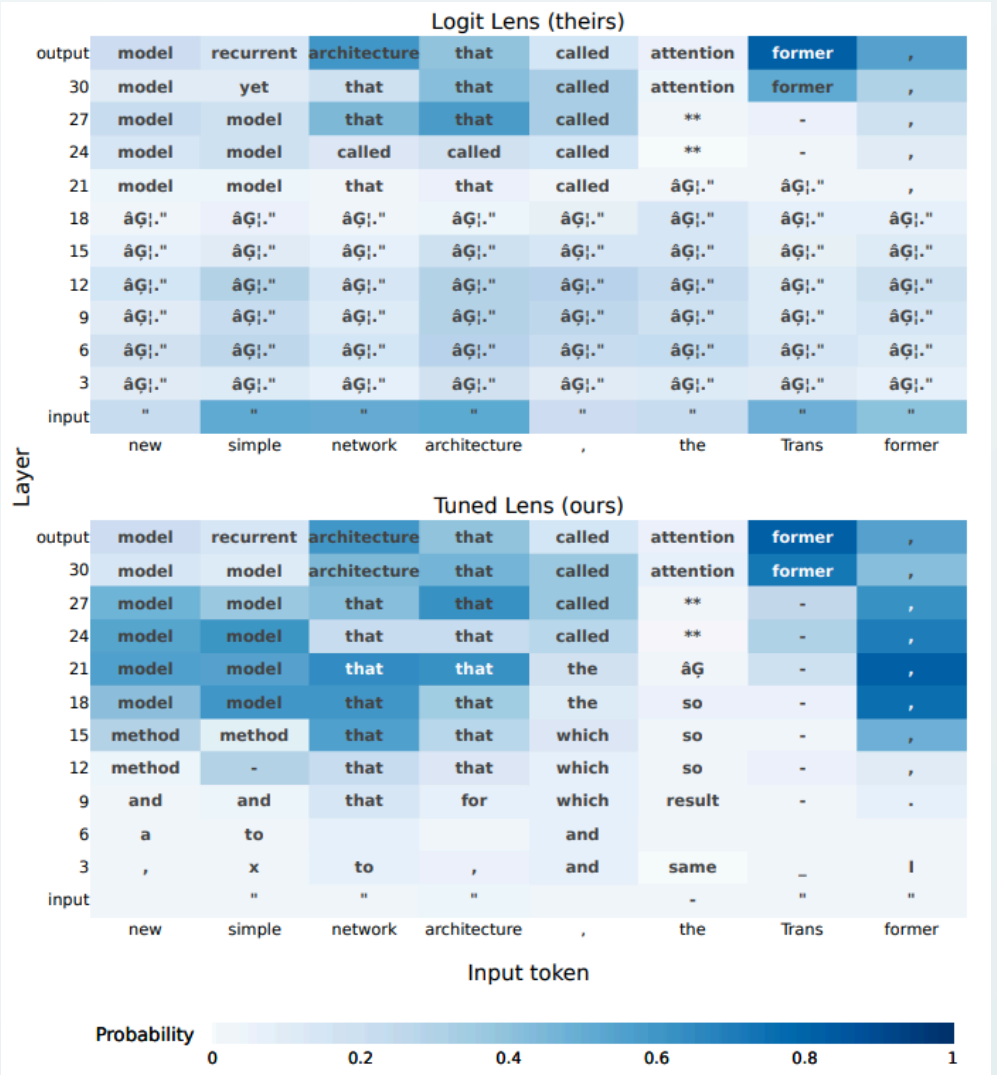
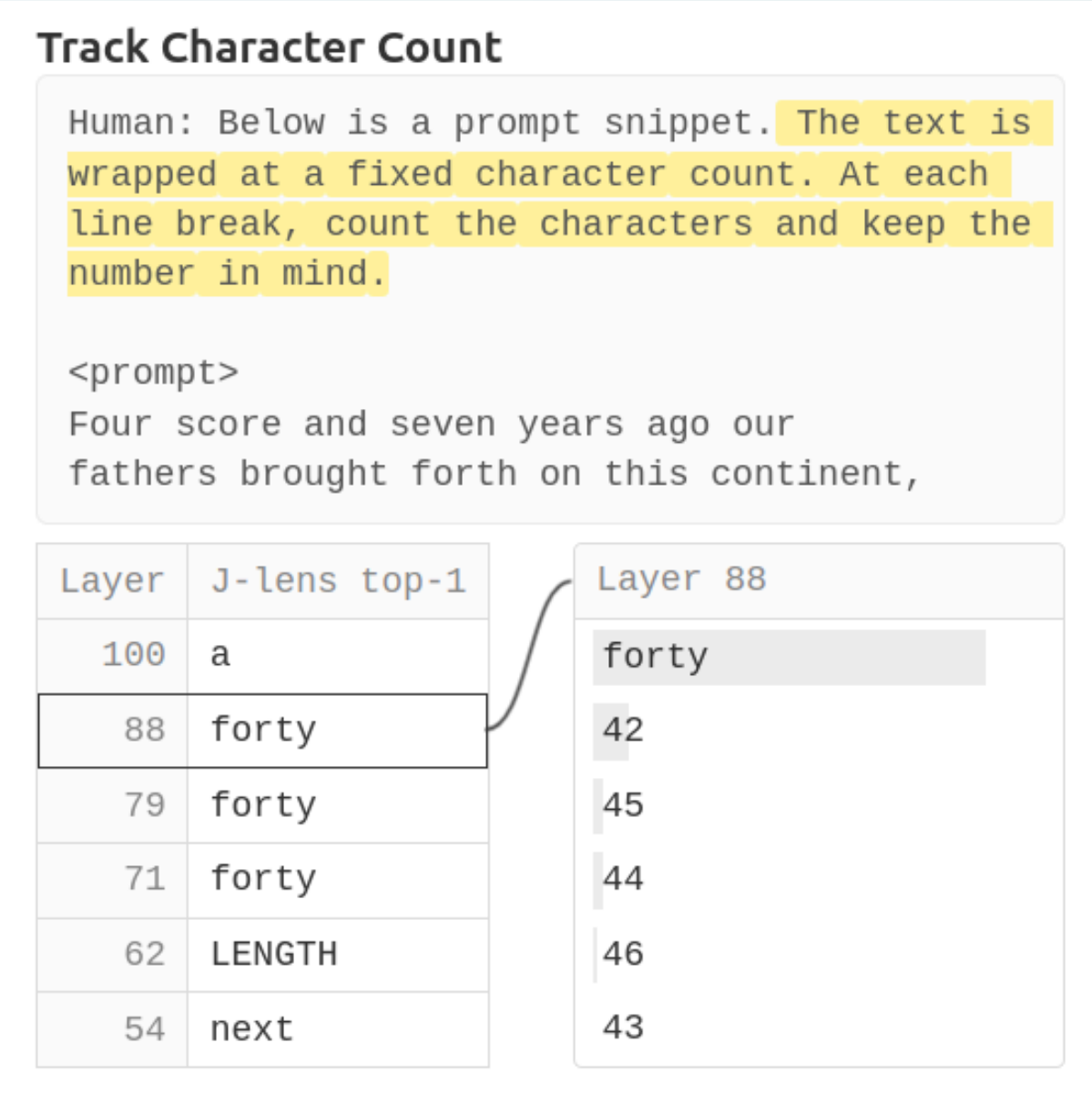



*Figure 1.* Comparison of our method, the *tuned lens* (bottom), with the "logit lens" (top) for GPT-Neo-2.7B prompted with an except from the abstract of Vaswani et al. (2017). Each cell shows the top-1 token predicted by the model at the given layer and token index. The logit lens fails to elicit interpretable predictions before layer 21, but our method succeeds.

# What can logitlenses++ tell us?

- Exploratory method
- Human-readable intermediate layers
- Confirm intuitions that model is building up predictions layer-wise

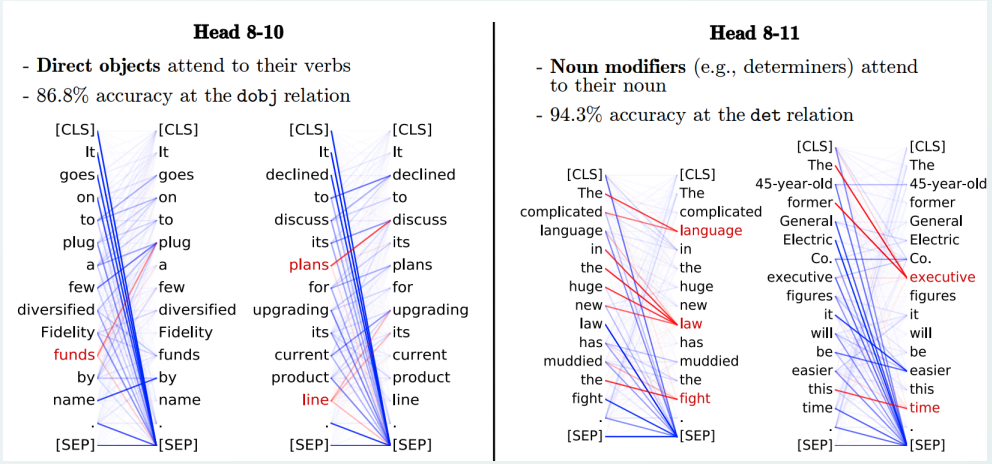


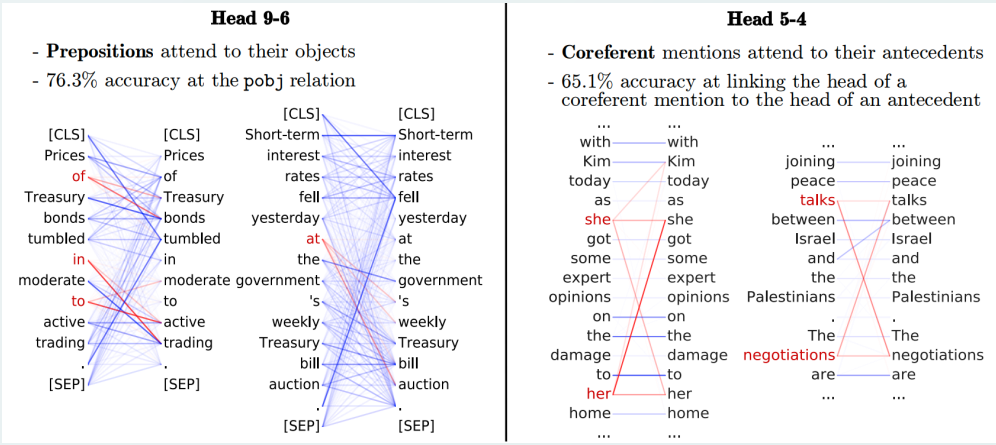
# Attention Patterns

Intuition:

- Transformers use attention to focus on important features/inputs.
- By tracking attention, we understand what the model considers important.

What Does BERT Look at? An Analysis of BERT's Attention (Clark et al., BlackboxNLP 2019)



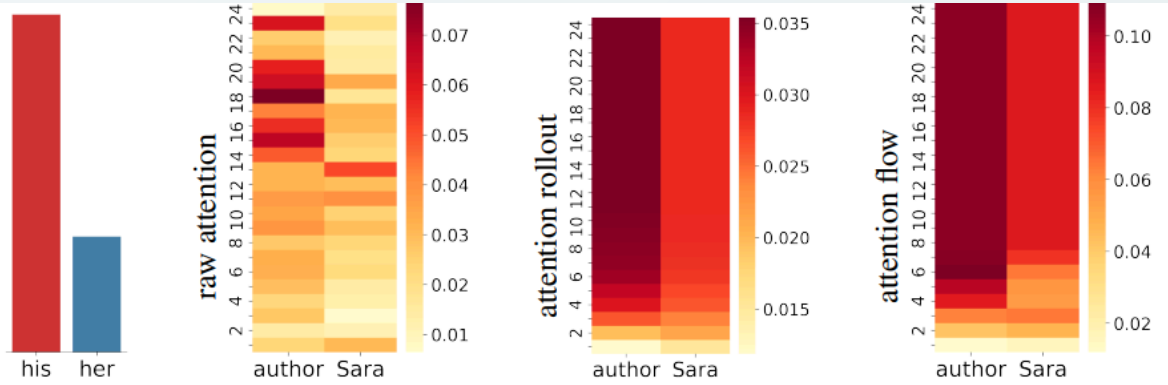



# Attention Flows

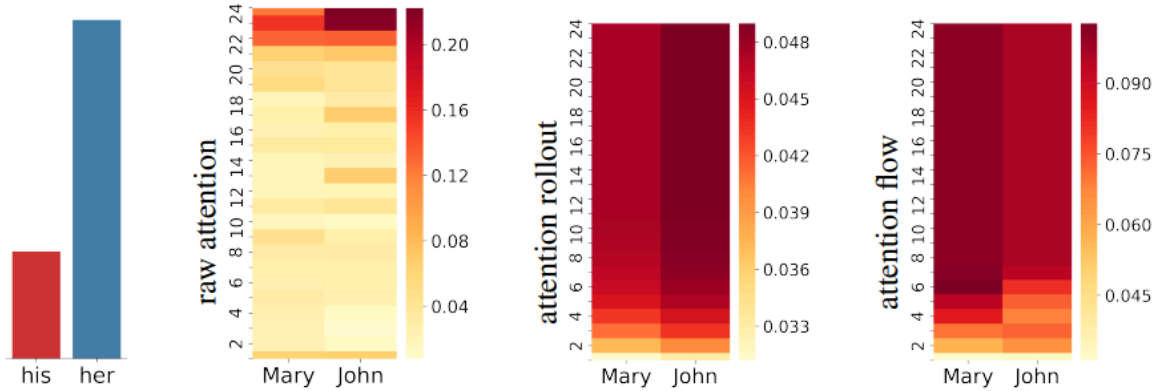
Attention is summation of previous attention layers: flow & rollout

Quantifying Attention Flow in Transformers (Abnar & Zuidema, ACL 2020

Goal: More faithful explanation




(a) "The author talked to Sara about mask book"




(b) *"Mary convinced John of* mask *love"*

Figure 4: Bert attention maps. We look at the attention weights from the mask embedding to the two potential references for it, e.g. "author" and "Sara" in (a) and "Mary" and "John" in (b). The bars, at the left, show the relative predicted probability for the two possible pronouns, "his" and "her".

# What can Attention tell us?

Have you ever looked at attention patterns?

Discuss!

# Attention as Explanation

Attention is not Explanation (Jain & Wallace, NAACL 2019)

- Not Consistent: Attention doesn't correlate with other feature importance measures
- Not Unique: Can find other attention patterns that don't change the prediction
    - This is a causal-counterfactual methodology: more next time!

Attention is not not Explanation (Wiegreffe & Pinter, EMNLP-IJCNLP 2019)

- "*An* explanation; not *the* explanation"
- Explanations that are faithful (to model) vs plausible (to humans)

Is Attention Explanation? An Introduction to the Debate (Bibal et al., ACL 2022)

# Extra: links forwards

Attention flows (not weights) are related to a *feature attribution* method (Shapley values)

Attention Flows are Shapley Value Explanations (Ethayarajh & Jurafsky, ACL-IJCNLP 2021)

Argument that *saliency* is better than attention for understanding:

The elephant in the interpretability room: Why use attention as explanation when we have saliency methods? (Bastings & Filippova, BlackboxNLP 2020)

# Today

**Descriptive** methods for looking into black box models

- exploratory analyses

# Next time

**Causal** models for understanding black box models.

http://en.wikipedia.org/wiki/Correlation_does_not_imply_causation

... but it's a start.